Exam Date & Time: 02-Jan-2025 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH MAKE UP EXAMINATIONS, JAN 2025

SIGNALS and SYSTEMS [BME 2125]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates: Answer ALL questions. Missing data may be suitably assumed.

1) Using discrete-time Fourier transform, compute $x_1(n) * x_2(n)$ when,

A) $x_1(n) = \{1, -2, 1\}$ (↑) (3)

and

$$x_2(n) = \begin{cases} 1, & 0 \le n \le 5 \\ 0, & elsewhere \end{cases}.$$

B) Interpret the statement - “DTFT is a special case of Z-transform”. (2)

C) The input $x(n)$ and the impulse response $h(n)$ of a system is given by

$$x(n) = \propto^n u(n)$$
&
$$h(n) = \beta^n u(n)$$

$$\propto \neq \beta$$ (5)

Compute the output $y(n)$ using Z-transform.

2) A) The ROC boundaries of a sequence $x(n)$ multiplied by an exponential sequence a^n, are multiplied by $|a|$. Illustrate with suitable examples. (3)

B) Consider an LSI system described by the difference equation,

$$y(n) - \frac{5}{6}y(n-1) + \frac{1}{6}y(n-2) = \frac{1}{3}x(n-1)$$ (3)

Determine the frequency response, $H(w)$, and impulse response, $h(n)$ of the system.

C) Analyze the discrete-time system, $y(n) = x(n)\,x(n-1)$ and find out (with justification) whether the system is: (4)

(i) Linear
(ii) Shift-invariant
(iii) Causal
(iv) Stable

3)

A) A discrete-time system has a unit sample response $h(n)$ given by

$$h(n) = \frac{1}{2}\delta(n) + \delta(n-1) + \frac{1}{2}\delta(n-2)$$

(3)

(i) Determine the frequency response $H(w)$ of the system.

(ii) Determine and sketch the magnitude response $|H(w)|$ over $-\pi \leq w \leq +\pi$.

B) "The ROC of a finite-length sequence is at least $0 < |z| < \infty$ and it may include either $z = 0$ or $z = \infty$." Illustrate this with an example. (3)

C) Analyze the following linear-shift invariant system with the impulse response,

$$h(n) = \begin{cases} a^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

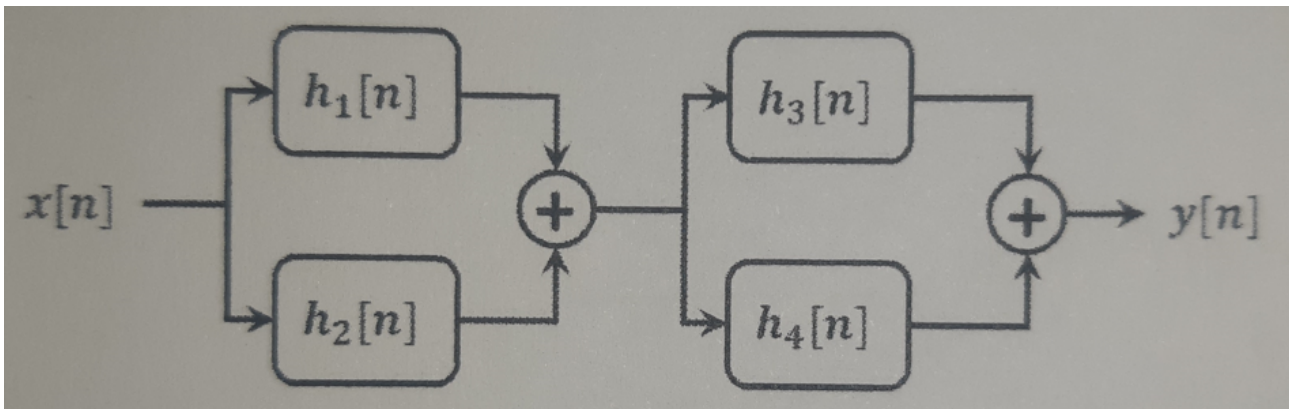
(4)

and identify (with justification) the range of the parameter $'a'$ for which the system is: (i) Stable (ii) Causal

4) Solve $x(n) * \delta(n)$ using both time-domain and frequency-domain methods. (4)

A)

B) Consider the interconnection of LSI systems, as shown in Fig. 1.



(4)

Fig. 1

(i) Determine the overall impulse response in terms of $h_1[n]$, $h_2[n]$, $h_3[n]$, and $h_4[n]$ when

$$h_1[n] = u[n]; h_2[n] = -u[n-1]; h_3[n] = u[n-1]; h_4[n] = \delta[n]$$

(ii) Determine the response $y[n]$ of the overall system for a given input $x[n] = \delta[n]$.

C) Determine the response of the following system,

$$y(n) = \frac{1}{3}[x(n+1) + x(n) + x(n-1)]$$

to the input signal $x(n)$ given below. (2)

$$x(n) = \begin{cases} |n|; & -3 \le n \le 3 \\ 0; & Otherwise \end{cases}$$

5) Consider a sequence,

A)
$$x(n) = \begin{cases} \left(\frac{1}{3}\right)^n, & n \ge 0 \\ \left(\frac{1}{2}\right)^{-n}, & n < 0 \end{cases}$$
(3)

(i) Indicate the characteristics of $x(n)$.

(ii) Based on the characteristics of $x(n)$, identify the ROC of the sequence without explicitly calculating the Z-transform.

B) Consider the difference equation of a moving average filter,

$$y(n) = \frac{1}{3}[x(n) + x(n-1) + x(n-2)]$$

(i) Determine the system's impulse response. (4)

(ii) Determine the frequency response $H(w)$, and plot $|H(w)|$ for the system over $0 \le w \le +\pi$.

C) Analyze and find out (with justification) whether the LSI system,

$$h(n) = \cos\left(\frac{\pi}{8}n\right)\{u(n) - u(n-10)\}$$
(3)

is both Causal and Stable.

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